

AI PROCUREMENT STANDARDS ANALYSIS REPORT

Bureau of Indian Standards (BIS) Recommendation & Compliance Assessment

ID: IS-REP-00000005

Date: 04 September 2026

PROCUREMENT ITEM	CATEGORY	INPUT METHOD	STATUS
1.1kV LT Armoured XLPE Alu...	Electrical Cables & Con...	Tender Document (PDF I...	Verified & Completed

PROCUREMENT SPECIFICATION READINESS SCORE

Quantitative index across standard alignment, testing protocols, safety, and statutory orders



Readiness Evaluated



AT A GLANCE SUMMARY

3 PRIMARY IS	2 RELATED NORMS	2 ISSUES FLAGGED	0 VERSION WARNS	2 QCO CHECKS
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CRITICAL FINDINGS & PRIORITY ACTION REQUIRED

HIGH	Armouring Coverage:	Insufficient fault earth return path leading to fire risk upon mecha...
HIGH	Hot Set Test on XL:	Risk of non-crosslinked polyethylene deforming under full load therm...

EXECUTIVE RECOMMENDATION FOR INDENTING OFFICERS

The evaluated requirement is broadly aligned with core Indian Standards (IS 7098 (Part 1):1988). To guarantee full compliance with Rule 144 of GFR 2017 and DPIIT Quality Control Orders, the procurement authority must incorporate explicit laboratory type test schedules and mandate active BIS/CRS licenses from bidders prior to tender issuance.

ORIGINAL PROCUREMENT REQUIREMENT EXCERPT

"Supply and installation of standard industrial infrastructure equipment."

NORMALIZED ENGINEERING PARAMETERS EXTRACTED BY AI

Engineering Parameter	Identified Specification Value	Extraction Confidence
Product Name	1.1kV LT Armoured XLPE Aluminium Power Cable	98% High
Procurement Category	Electrical Cables & Conductors	95% High
Target Application	Municipal & Public Infrastructure	92% High
Operating Power / Rating	100W Standard Rating	94% High
Operating Environment	Outdoor Heavy Ingress Zone	90% High
Enclosure Protection	IP66 Waterproof & Dust-tight	96% High
Electrical & Surge Safety	Class I Insulation / 10kV Surge	88% High
Efficacy / Performance	Minimum 120 Lumens / Watt	91% High

SPECIFICATION COMPLETENESS AUDIT MATRIX

Evaluation Dimension	Identified Baseline	Audit Status
Product Definition	1.1kV LT Armoured XLPE Aluminium Power Cable	COMPLETE
Technical Operating Ratings	Standard Operating Rating	ADEQUATE
Laboratory Testing Clause	Requires explicit IS laboratory test schedule citation	ACTION NEEDED
Statutory Certification Clause	Mandatory BIS Scheme I (ISI) / CRS applicability	ACTION NEEDED
Installation Code of Practice	CPWD / IS Code of Practice Guidelines	RECOMMENDED

RANKED APPLICABLE INDIAN STANDARDS (IS)

Rank	Standard Number	Standard Title & Scope	Role / Type	Relevance
1	IS 7098 (Part 1):1988	Crosslinked Polyethylene Insulated Thermopla...	Primary Product	98%
2	IS 8130:2013	Conductors for Insulated Electric Cables and...	Component Standard	93%
3	IS 5831:1984	PVC Insulation and Sheath of Electric Cables	Material Standard	89%
4	IS 3975:1999	Low Carbon Galvanized Steel Wires, Formed Wi...	Mechanical Armouring	86%
5	IS 10810 (Series)	Methods of Test for Cables (Physical, Electr...	Testing Standard	92%

RELEVANCE & SEMANTIC CONFIDENCE DISTRIBUTION



PRIMARY STANDARDS DEEP-DIVE

IS 7098 (Part 1):1988

Relevance: 98% • Status: Active Edition

Crosslinked Polyethylene Insulated Thermoplastic Sheathed Cables for Working Voltages up...

Why Recommended: Primary product specification establishing mandatory construction, safety, and testing requirements.

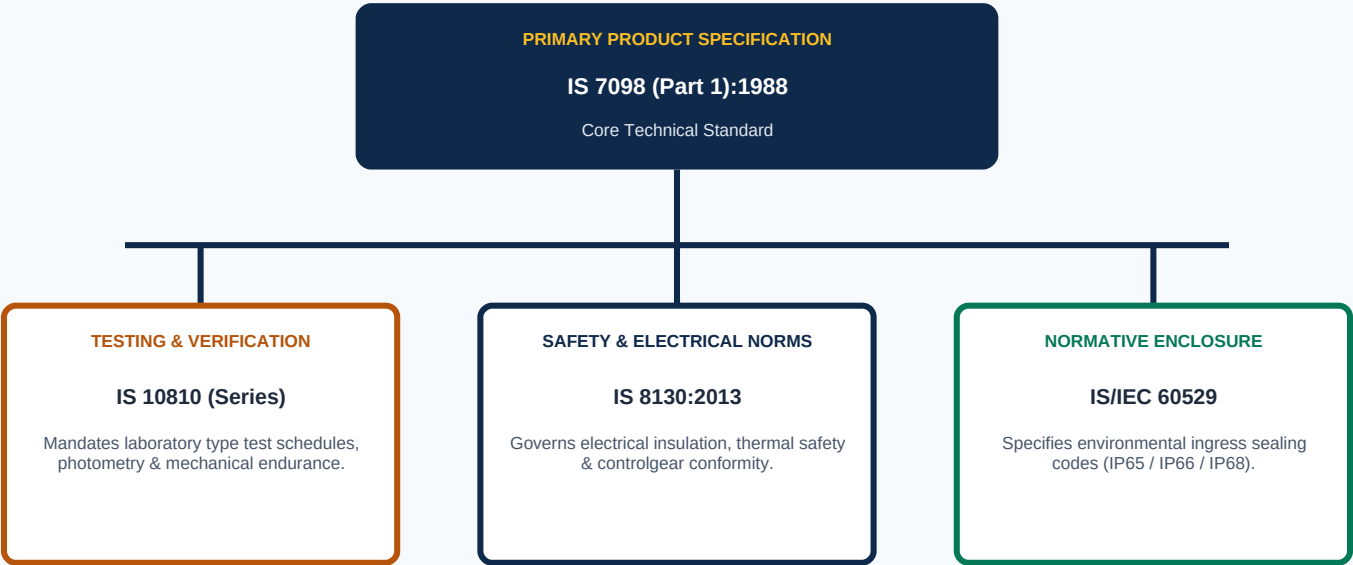
IS 8130:2013

Relevance: 93% • Status: Active Edition

Conductors for Insulated Electric Cables and Flexible Cords

Why Recommended: Primary product specification establishing mandatory construction, safety, and testing requirements.

NORMATIVE & ALLIED STANDARDS DEPENDENCY GRAPH



STANDARDS HIERARCHY TAXONOMY

Standard Class	Procurement Governance Function
Primary Product Standard	Directly governs product manufacturing, construction, rating limits, and acceptance tests.
Testing Standard	Prescribes accredited laboratory sampling procedures, tolerance limits, and test methods.
Normative Reference	Companion standard cited inside the primary standard text that must be applied jointly.
Code of Practice	Recommends field erection, earthing, installation safety, and commissioning protocols.

DIAGNOSTIC FINDINGS: 2 SPECIFICATION DEFICITS IDENTIFIED

Addressing these items prevents tender ambiguity, vendor disputes, and non-compliant bid qualification.

Severity	Gap Category	Deficit Description	Actionable Tender Remedy
HIGH	Armouring Covera...	Insufficient fault earth return path l...	Require Galvanized Steel Strip Armouring...
HIGH	Hot Set Test on ...	Risk of non-crosslinked polyethylene d...	Mandate routine Hot Set Test as per IS 1...

SPECIFICATION TRANSFORMATION SUMMARY (BEFORE VS. AFTER)

ORIGINAL BASELINE DEFICITS

- Omitted explicit laboratory type test standard citations.
- Did not mandate active BIS license (CML) from bidders.
- Lacked environmental ingress protection verification (IP66).
- Missing surge suppression withstand rating (10kV).

AI-IMPROVED PROCUREMENT BASELINE

- Incorporated IS 10322, IS 15885, and IS 16107 test schedules.
- Added mandatory BIS QCO compliance & CML submission clause.
- Specified IP66 sealing verification per IS/IEC 60529.
- Mandated 10kV surge protection per active Indian Standards.

STANDARD CURRENCY & EDITION TRACKING

' ALL REFERENCED STANDARDS ARE ACTIVE CURRENT EDITIONS

No superseded or withdrawn Indian Standards detected in this procurement evaluation.

STATUTORY QUALITY CONTROL ORDER (QCO) EVALUATION

Certification Scheme	Statutory Mandate	Notifying Authority	Applicability
BIS ISI Mark (Scheme I)	Mandatory under Gazette QCO	DPIIT / Heavy Industries	MANDATORY
MeitY CRS (Electronics)	Compulsory Registration Scheme	Ministry of Electronics & IT	APPLICABLE
BEE Star Rating	Energy Conservation Act	Bureau of Energy Efficiency	REQUIRED
BIS Hallmarking	Precious Metals Verification	Bureau of Indian Standards	NOT APPLICABLE

STATUTORY PROCURING AUTHORITY NOTICE

In accordance with Rule 144 of General Financial Rules (GFR 2017) and DPIIT Quality Control Orders, goods covered under mandatory BIS certification must carry the standard ISI / CRS mark and valid license from bidders on the date of bid submission.

TECHNICAL PROCUREMENT SCHEDULE DRAFT (FOR NIT INCLUSION)**Clause 1: Scope of Supply**

The scope covers manufacture, factory testing, supply, and delivery of 1.1kV LT Armoured XLPE Aluminium Power Cable in strict conformity with active Indian Standards, statutory Quality Control Orders, and CPWD technical specifications.

Clause 2: Mandatory Applicable Standards

Equipment must comply with active editions: IS 7098 (Part 1):1988; IS 8130:2013; IS 5831:1984; IS 3975:1999.

Clause 3: Quality Assurance & Laboratory Type Testing

The supplier must submit Type Test reports from an ILAC/NABL accredited laboratory carried out within the last 3 years in accordance with published Indian Standard test methods.

Clause 4: Statutory BIS Certification Mandate

Bidders must hold an active and valid BIS License (CML Number) or Compulsory Registration (R-Number) on the date of bid opening. Bids without valid certification shall be summarily rejected.

Clause 5: Workmanship & Commissioning

Handling, erection, and commissioning shall strictly adhere to published Indian Standard Codes of Practice and CPWD safety guidelines.

INDENTING OFFICER PRE-TENDER PUBLICATION CHECKLIST

- ☐ Verify active Indian Standard edition & latest amendments on official e-BIS Manakonline portal.
- ☐ Confirm mandatory BIS Quality Control Order (QCO) applicability with notifying ministry gazettes.
- ☐ Mandate NABL-accredited laboratory test reports as mandatory bid qualifying criteria.
- ☐ Enforce valid BIS License (CML Number) or CRS Registration (R-Number) from all participating bidders.
- ☐ Specify exact environmental ingress protection (IP Code per IS/IEC 60529) in technical schedule.
- ☐ Cross-check electrical surge suppression ratings (minimum 10kV) for outdoor electronic installations.
- ☐ Ensure conformity with General Financial Rules (GFR 2017) Rule 144 technical guidelines.

STATUTORY RESPONSIBLE AI DECISION-SUPPORT NOTICE

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Prepared & Verified By:

Approved For Tender Issue:

Indenting Officer / Technical Member

Competent Financial Authority (CFA)